

Lab 3: Architectural Pattern Selection — Written Justification

Online Bookstore System (Jackfruit Phase-1)

Team: Varchas (PES2UG24CS575) · Udit Mourya (PES2UG24CS565) · Vrinda Venugopal M (PES2UG24CS601)

Architecture Selection: We chose Layered Architecture for the Online Bookstore System.

Reason 1 — Natural separation of concerns

The bookstore's workflow already splits cleanly into three concerns: the customer-facing browsing/ordering interface, the business logic that manages orders and payments, and the persistent storage of the book catalog, user accounts, and order history. A layered architecture maps directly onto these boundaries, keeping each concern independently understandable and testable without the coordination overhead a more distributed style would add.

Reason 2 — Right-sized for the system's scale

Core functions such as search, cart management, and checkout are largely sequential and don't require the independent scaling or fault isolation that microservices are built for. A layered approach delivers the same modularity and replaceability of components (e.g. swapping the Payment Service) while avoiding the network latency and operational complexity of running and deploying multiple independent services for a system of this size.

Security Advantage

Routing all payment processing through a single Payment Service component in the business layer means the presentation layer never handles raw card data directly. Card details pass through one controlled interface (ProcessPayment) rather than being touched by multiple components, which narrows the attack surface for sensitive payment information.

Performance Benefit

Because the Catalog component queries the Database component directly within the same in-process layered stack, read-heavy operations like book search and browsing avoid the network round-trips that a distributed, independently-deployed service architecture would introduce — keeping response times low for the most frequent user action in the system.